KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

(KIIT University)

DATA FUNDAMENTALS PROJECT REPORT

Student Performance Analysis

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Subject : Data Fundamentals

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1. Problem Statement (Original Question)

A university collects academic performance data for students enrolled in different departments such as Computer Science and Information Technology. The dataset contains marks obtained by students in subjects like Mathematics, Science, and Programming along with attendance records.

The academic administration wants to evaluate student performance to understand learning outcomes and identify students who may require academic support.

The university wants to answer the following questions:

- Who are the top-performing students?
- Which department performs better academically?
- Are students with higher attendance performing better?
- Which students are at risk due to low marks?

Students will analyze the dataset and generate insights to help the academic committee improve teaching strategies.

SQL Tasks

Students should:

- Display student records.
- Calculate average marks of each student.
- Identify top-performing student.
- Calculate department-wise average marks.
- Identify students scoring below 60 marks.
- Calculate overall class average.

Excel Analysis Tasks

Students must:

- Create column Average Marks.
- Sort students by highest average marks.
- Apply conditional formatting to highlight marks below 60.
- Create Pivot Table (Rows → Department, Values → Average Marks).

Python Analysis Tasks

Students must:

- Load dataset using Pandas.
- Create column `average_marks`.

- Identify top-performing student.
- Group data by department.
- Calculate average marks by department.
- Analyze relationship between attendance and marks.

Visualization Tasks

Students should create:

- Bar chart → Student average marks
- Column chart → Department performance
- Scatter plot → Attendance vs marks

2. Dataset Overview

The dataset used is `student_dataset_1000.csv`, containing records of 1000 students across multiple engineering departments.

Column Name	Data Type	Description
student_id	Integer	Unique student identifier
name	String	Student full name
department	String	Department (CS, IT, ECE, EEE, ME, Civil)
math	Integer	Marks in Mathematics (0–100)
science	Integer	Marks in Science (0–100)
programming	Integer	Marks in Programming (0–100)
attendance	Integer	Attendance percentage (0–100)

3. SQL Analysis

SQLite was used in-memory via Python's `sqlite3` library. The CSV dataset was loaded into a table named `students` and the following queries were executed.

3.1 Display Student Records

Task: Display student records.

```
SELECT * FROM students LIMIT 6;
```

This query retrieves the first 6 student records to verify that the data was loaded correctly into the in-memory SQLite database.

3.2 Average Marks of Each Student

Task: Calculate average marks of each student.

```
SELECT student_id, name,
       (math + science + programming)/3.0 AS avg_marks
FROM students;
```

This computes the average of the three subjects for every student and displays it alongside their ID and name.

3.3 Top-Performing Student

Task: Identify top-performing student.

```
SELECT student_id, name,
       (math + science + programming)/3.0 AS avg_marks
FROM students
ORDER BY avg_marks DESC
LIMIT 1;
```

This orders all students by their average marks in descending order and returns only the top student. **Pallavi Nair** (Student ID: 176, Civil department) scored the highest with 100 marks in all three subjects.

3.4 Department-wise Average Marks

Task: Calculate department-wise average marks.

```
SELECT department,
       AVG((math + science + programming)/3.0) AS dept_avg
FROM students
GROUP BY department;
```

Civil department had the highest average (≈72.88), while EEE had the lowest (≈70.54).

3.5 Students Scoring Below 60

Task: Identify students scoring below 60 marks.

```
SELECT * FROM students
WHERE math < 60 OR science < 60 OR programming < 60;
```

This identifies all at-risk students who scored below 60 in at least one subject. These students may require additional academic support.

3.6 Overall Class Average

Task: Calculate overall class average.

```
SELECT AVG((math + science + programming)/3.0) AS overall_avg
FROM students;
```

The overall class average across all departments and subjects was approximately **71.37 marks**.

3.7 Count Students in Each Department

(Additional Query)

```
SELECT department, COUNT(*) AS total_students
FROM students
GROUP BY department;
```

This provides the enrollment count per department, giving useful context when comparing department-level averages.

3.8 Students with Attendance Above 90

(Additional Query)

```
SELECT student_id, name, attendance
FROM students
WHERE attendance > 90;
```

Used to identify high-attendance students as a preliminary step before the correlation analysis between attendance and performance.

3.9 Highest Marks in Each Subject

(Additional Query)

```
SELECT
    MAX(math)        AS max_math,
    MAX(science)     AS max_science,
    MAX(programming) AS max_programming
FROM students;
```

All three subjects had a maximum score of 100, confirming that full marks were achievable and some students did achieve them.

3.10 Students Above Class Average

(Additional Query)

```
SELECT student_id, name,
       (math + science + programming)/3.0 AS avg_marks
FROM students
WHERE (math + science + programming)/3.0 >
      (SELECT AVG((math + science + programming)/3.0)
       FROM students);
```

This uses a subquery to identify students who scored above the overall class average. Such students can be considered for scholarships or advanced academic programs.

4. Excel Analysis

The Excel file `DATA_FUNDAMENTAL_PROJECT.xlsx` contains three sheets: `ORIGINAL_DATASET`, `OUTPUT`, and `PIVOT TABLE`.

4.1 Average Marks Column

Task: Create column Average Marks.

A new column named **Average Marks** was added using the Excel formula: `=AVERAGE(Math, Science, Programming)` for each student row. This computes the mean score across all three subjects per student.

4.2 Sort by Highest Average Marks

Task: Sort students by highest average marks.

The dataset was sorted in descending order on the Average Marks column. Pallavi Nair (100 in all subjects) appeared at the top of the sorted list.

4.3 Conditional Formatting

Task: Apply conditional formatting to highlight marks below 60.

Conditional formatting rules were applied to the Math, Science, and Programming columns. Any cell containing a value below 60 was highlighted to visually flag students at academic risk.

4.4 Pivot Table

Task: Create Pivot Table (Rows → Department, Values → Average Marks).

A Pivot Table was created to summarize department-wise performance. Results:

Department	Average of Average Marks
Civil	72.88
CS	71.58
ECE	70.88
EEE	70.54
IT	71.34
ME	71.16
Grand Total	**71.37**

Civil department had the highest average (72.88) and EEE had the lowest (70.54).

5. Python Analysis

5.1 Load Dataset

Task: Load dataset using Pandas.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('student_dataset_1000.csv')
df.head()
df.info()
```

```
df.describe()
```

The dataset was loaded into a Pandas DataFrame. `df.info()` confirmed 7 columns with no missing values, and `df.describe()` provided summary statistics.

5.2 Create average_marks Column

Task: Create column average_marks.

```
df['Average_Marks'] = (df['math'] + df['science'] + df['programming'
   ]) / 3
df['Total_Marks']   = df['math'] + df['science'] + df['programming']
```

Two new columns were created: `Average_Marks` (mean of three subjects) and `Total_Marks` (sum of all three subjects).

5.3 Identify Top-Performing Student

Task: Identify top-performing student.

```
top_student = df.loc[df['Total_Marks'].idxmax()]
print("Top Performing Student:")
print(top_student)
```

The student with the maximum `Total_Marks` was identified as **Pallavi Nair** (Student ID: 176, Civil department) with 300 total marks (100 in each subject).

5.4 Group Data by Department

Task: Group data by department and calculate average marks by department.

```
dept_summary =
    df.groupby('department').agg({ 'math': 'mea
    n',
    'science': 'mean', 'p
    rogramming': 'mean', '
    attendance': 'mean',
    'Total_Marks': 'mean',
     'student_id': 'count
    '
}).rename(columns={ 'ma
    th': 'avg_math',
    'science': 'avg_science', 'programmin
    g': 'avg_programming', 'attendance':
    'avg_attendance', 'Total_Marks': 'avg
    _total_marks', 'student_id': 'total_s
```

The `groupby` operation produced a summary table showing per-department averages for each subject, attendance, and total marks, along with the student count.

5.5 Attendance vs Marks Relationship

Task: Analyze relationship between attendance and marks.

```
correlation = df[['attendance', 'math', 'science',
                  'programming', 'Total_Marks']].corr()
print("Correlation Matrix:")
print(correlation)
```

The correlation between attendance and Total Marks was low (close to 0), indicating that attendance alone is not a strong predictor of academic performance.

6. Visualizations

6.1 Bar Chart – Student Average Marks

```
plt.figure(figsize=(12, 5))
plt.bar(df['name'], df['Average_Marks'])
plt.title("Student Average Marks")
plt.xlabel("Student")
plt.ylabel("Average Marks")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

A bar chart was created showing each student's average marks. It helped visualize the distribution of performance and identify high and low performers across the class.

6.2 Column Chart – Department Performance

```
dept_perf = df.groupby('department')['Total_Marks'].mean().
   reset_index()

plt.figure(figsize=(8, 5))
sns.barplot(x='department', y='Total_Marks', data=dept_perf)
plt.title("Department Performance (Average Total Marks)")
plt.xlabel("Department")
plt.ylabel("Average Total Marks")
plt.show()
```

The column chart showed Civil department as the highest-performing department, with all departments scoring within a narrow range of approximately 70–73 marks.

6.3 Scatter Plot – Attendance vs Marks

```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='attendance', y='Total_Marks', data=df)
plt.title("Attendance vs Total Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("Total Marks")
plt.show()
```

The scatter plot showed a wide, random spread with no clear linear trend, confirming the weak correlation between attendance and marks.

6.4 Heatmap – Correlation Matrix (Additional)

```
plt.figure(figsize=(8, 6))
corr =
           df[['math', 'science', 'programming'
           , 'attendance', 'Total_Marks']].corr(
           )
sns.heatmap(corr, annot=True, cmap='coolwarm')
```

The heatmap confirmed that math, science, and programming marks are moderately correlated with each other, while attendance shows weak correlation with academic performance.

6.5 Boxplot and Pairplot (Additional)

```
# Boxplot
plt.figure(figsize=(10, 5))
sns.boxplot(data=df[['math', 'science', 'programming', 'attendance'
   ]])
plt.title("Boxplot of Features")
plt.show()

# Pairplot
sns.pairplot(df[['math', 'science', 'programming', 'attendance']])
plt.show()
```

The boxplot showed no extreme outliers in any feature. The pairplot revealed that subject marks have a positive pairwise relationship, while attendance is relatively independent of the other variables.

7. Insights and Findings

- **Top Performer:** Pallavi Nair (Student ID: 176, Civil Department) scored 100 marks in all three subjects — the highest in the dataset.
- **Department Performance:** Civil department had the highest average marks (≈72.88), while EEE had the lowest (≈70.54). Differences between departments were small (within ≈2.3 marks).
- **At-Risk Students:** Several students scored below 60 in at least one subject. These students need targeted academic support and remedial classes.
- **Attendance vs Performance:** The correlation between attendance and total marks was weak, suggesting that attending class alone does not guarantee high marks. Study habits and conceptual understanding may play a larger role.
- **Class Average:** The overall class average was approximately 71.37 out of 100, indicating a moderate-to-good level of academic performance.

- **Subject Correlation:** Math, Science, and Programming marks were moderately correlated, meaning students who perform well in one subject tend to perform reasonably in others.
- **Above-Average Students:** A significant portion of students performed above the class average, which is a positive indicator for the university's academic standards.

8. Conclusion

This project involved a comprehensive analysis of student academic performance data using SQL, Excel, and Python. The three tools complemented each other — SQL was used for querying and filtering data, Excel provided visual formatting and pivot summaries, and Python enabled deeper statistical analysis and visualizations.

The analysis revealed that while most students perform at a moderate-to-good level, a subset of students are at risk of academic failure due to low subject marks. The weak link between attendance and marks is an important finding that suggests the university should focus not just on attendance policies but also on the quality of learning and student understanding.

These insights can help the academic committee design better intervention programs, identify students for scholarships or awards, and improve overall teaching strategies across departments.

— End of Report —

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